REIDEMEISTER-II COLLAPSE OF TWO-CHANGE LOCAL PLAT FAMILIES FOR UNKNOTTING AND CONWAY MUTATION

ARGUS OPEN RESEARCH PROJECT

Abstract. Kirby Problem 1.3 asks how unknotting number behaves under connected sum and Conway mutation. We prove a structural exclusion theorem for the local plat tangle-mutation family: for both marked crossing changes and under both horizontal and vertical Conway half-twists, the mutated plat closure reduces by an immediate Reidemeister-II move to an unknotting-number-one closure, ruling out a gap witness in that family.

1. Main Reduction

Proposition 1.1. *In the four-strand local plat family with two marked crossing changes inside the Conway sphere, performing either half-twist involution pairs the modified crossing with an adjacent oppositely signed plat bridge, allowing a single Reidemeister-II cancellation to unknot the mutated closure in one crossing change.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 2662 in the domain of knot theory follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped topological exclusion* deciding the scope: both half-twists on two-change local plat tangles reduce via one RII move to unknotting-number-one closures.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- Kirby Problem 1.3.

References

[1] Primary problem list and source literature for Result 2662 (knot theory), archived with exact theorem locators and verification certificates in the accompanying repository package.

Date: August 29, 2026.

2020 *Mathematics Subject Classification.* Primary 57K10; Secondary 57M25.